

VALIDATION REPORT

OPERATIONAL QUALIFICATION

STATISTICAL PROCESS CONTROL MODULE

JR Validated Environment — Statistical Process Control Module v2.0

Field	Value
Document Number	JR-VR-SPC-001
Title	Validation Report — Operational Qualification, Statistical Process Control Module
Validation Plan	JR-VP-SPC-001 v2.0
System	JR Validated Environment — Statistical Process Control Module
Module Version	2.0
Document Version	2.0
Execution Date	2026-05-18
OQ Result	PASS — 83 / 83 tests passed, 0 failures, 0 deviations
Author	Joep Rous
Reviewer	[Name]
Approver	[Name]

Version History

Version	Date	Author	Description
1.0	2026-03-31	Joep Rous	Initial OQ execution report. All 71 test cases passed. No deviations.
2.0	2026-05-18	Joep Rous	Updated for SPC module v2.0. Added --report PV report TCs for jrc_spc_imr, jrc_spc_xbar_r, jrc_spc_xbar_s, and jrc_spc_p (IMR-016..018, XBR-017..019, XBS-016..018, P-014..016). 83 / 83 tests passed.

Approval Signatures

Role	Name	Signature	Date
Author	Joep Rous		2026-05-18
Reviewer	[Name]		
Approver (QA)	[Name]		

1. Purpose

This document reports the execution and outcome of the Operational Qualification (OQ) test suite for the JR Validated Environment Statistical Process Control Module, version 2.0. It provides objective evidence that all SPC scripts perform as specified in the OQ Validation Plan JR-VP-SPC-001 v2.0 under the defined test conditions.

This report records the test environment, execution results for all 83 test cases, and the formal OQ conclusion. Results are derived directly from the evidence file generated by `admin_spc_oq`.

2. Scope

2.1 In Scope

- All 5 R scripts in `repos/spc/R/` (`jrc_spc_imr`, `jrc_spc_xbar_r`, `jrc_spc_xbar_s`, `jrc_spc_p`, `jrc_spc_c`)
- Shewhart control chart computations with 8 Western Electric rules
- PNG output to `~/Downloads/`
- User-specified control limit override (`--ucl` / `--lcl` flags)
- Correct rejection of invalid inputs with informative error messages
- Bypass-protection verification (`RENV_PATHS_ROOT` check)
- Numeric correctness: control limits verified against independently computed reference values using Shewhart constants ($d_2=1.128$, $D_4=3.267$)
- OOC observation flagging — flagged subgroup or individual identified with rule number
- IN CONTROL / OUT OF CONTROL verdict for stable and out-of-control datasets

2.2 Out of Scope

- Infrastructure scripts in `bin/` and `admin/` (covered by JR-IQ-001)
- Performance qualification (PQ) with real production data
- Validation of R interpreter and third-party packages (`ggplot2`, `qcc`)
- Statistical validation of the underlying control chart theory — method references are cited in JR-VP-SPC-001 Section 3

3. Reference Documents

Document ID	Title	Location
JR-VP-SPC-001 v2.0	Validation Plan — OQ, Statistical Process Control Module	<code>repos/spc/docs/spc_validation_plan.pdf</code>
JR-VP-002 v1.0	Validation Plan — OQ, Community Script Suite v1.1.0	<code>docs/oq_validation_plan.docx</code>
JR-VP-001 v1.0	Validation Plan — Installation Qualification	<code>docs/</code>
JR-IQ-001	IQ Execution Evidence	<code>docs/iq_validation_20260311_205146.txt</code>
OQ Evidence	SPC OQ Execution Evidence (pytest output + header)	<code>spc_oq_execution_20260518T172901.txt</code>

4. Test Environment

Component	Details
Execution date / time	2026-05-18T17:29:01
Host	Joeps-MacBook-Air.local
Operating System	[not found]
R Version	4.6.0
Python Version	3.11.9
pytest Version	pytest 8.3.4
Test runner	repos/spc/admin_spc_oq
Test directory	repos/spc/oq/
Test data	repos/spc/oq/data/ (12 CSV fixtures)
Execution command	bash repos/spc/admin_spc_oq
OQ evidence file	spc_oq_execution_20260518T172901.txt

5. Test Data

All test data CSV files reside in repos/spc/oq/data/. Each file contains two columns: id and value (or id, subgroup, value for subgroup charts). Numeric reference values were computed independently and are used as acceptance criteria for numeric correctness test cases.

File	Description	Key Properties
imr_stable.csv	I-MR stable dataset	n=25 individuals; X-bar=10.0668, MR-bar=0.32125; no WE violations
imr_ooc.csv	I-MR out-of-control dataset	imr_stable with obs 13 = 11.50 (Rule 1 violation)
imr_missing_col.csv	I-MR missing column fixture	CSV with no 'value' column — triggers error handling
imr_one_obs.csv	I-MR too few observations fixture	CSV with only 1 row — triggers too-few-observations error
xbar_r_stable.csv	X-bar & R stable dataset	20 subgroups × n=5; grand X-bar=50.165; no WE violations
xbar_r_ooc.csv	X-bar & R out-of-control dataset	xbar_r_stable with sg09 mean≈53 (Rule 1 violation)
xbar_s_stable.csv	X-bar & S stable dataset	20 subgroups × n=8; grand X-bar=100.060; no WE violations
xbar_s_ooc.csv	X-bar & S out-of-control dataset	xbar_s_stable with sg09 mean≈102 (Rule 1 violation)
p_stable.csv	P-chart stable dataset	25 subgroups n=100 each; p-bar=0.04160; no WE violations
p_ooc.csv	P-chart out-of-control dataset	p_stable with subgroup 13 = 15 defectives (Rule 1 violation)
c_stable.csv	C-chart stable dataset	25 subgroups of defect counts; c-bar=4.960; no WE violations
c_ooc.csv	C-chart out-of-control dataset	c_stable with subgroup 12 = 15 defects (Rule 1 violation)

6. Test Results

All 83 test cases were executed on 2026-05-18. Results are derived from the OQ evidence file and reproduced below. The evidence file contains the complete pytest output including individual test IDs, pass/fail status, and timing.

6.1 Summary

Script	Tests Executed	Result
--------	----------------	--------

jrc_spc_imr	18	18/18 PASS
jrc_spc_xbar_r	19	19/19 PASS
jrc_spc_xbar_s	18	18/18 PASS
jrc_spc_p	16	16/16 PASS
jrc_spc_c	12	12/12 PASS
TOTAL	83	83/83 PASS

6.2 Evidence File

Item	Details
Filename	spc_oq_execution_20260518T172901.txt
Timestamp	2026-05-18T17:29:01
Host	Joeys-MacBook-Air.local
Full path	spc_oq_execution_20260518T172901.txt

6.3 jrc_spc_imr — 18/18 PASS

Script: jrc_spc_imr.R — I-MR (Individuals & Moving Range) chart. Test data: imr_stable.csv (stable), imr_ooc.csv (out-of-control).

TC ID	Description	Expected	Result
TC-SPC-IMR-001	Stable dataset — exit 0, control limit section and verdict present	Exit 0, control limits, stability section, verdict present	PASS
TC-SPC-IMR-002	Stable data — IN CONTROL verdict	Exit 0, 'IN CONTROL' or 'STABLE' in output	PASS
TC-SPC-IMR-003	OOC dataset — exit 0, OUT OF CONTROL in output	Exit 0, 'OUT OF CONTROL' or 'SIGNALS' in output	PASS
TC-SPC-IMR-004	OOC dataset — observation 13 flagged as Rule 1 violation	Exit 0, '13' and '[1]' or 'Rule 1' in output	PASS
TC-SPC-IMR-005	--ucl / --lcl flags accepted, user-specified limits reported	Exit 0, user-specified UCL mentioned	PASS
TC-SPC-IMR-006	PNG written to ~/Downloads/	PNG present, recent mtime	PASS
TC-SPC-IMR-007	No arguments — non-zero exit, usage message	Exit ≠ 0, 'Usage' in output	PASS
TC-SPC-IMR-008	File not found — non-zero exit	Exit ≠ 0, 'not found' in output	PASS
TC-SPC-IMR-009	Missing 'value' column — non-zero exit, column named	Exit ≠ 0, 'value' in output	PASS
TC-SPC-IMR-010	Too few observations (1 row) — non-zero exit	Exit ≠ 0	PASS
TC-SPC-IMR-011	Bypass protection — direct Rscript call fails	Exit ≠ 0, 'RENV_PATHS_ROOT' in output	PASS
TC-SPC-IMR-012	X-bar = 10.0668 ± 0.0001 (imr_stable.csv)	X-bar: 10.0668 ± 0.0001	PASS
TC-SPC-IMR-013	UCL_I = 10.9212 ± 0.001 (imr_stable.csv)	UCL: 10.9212 ± 0.001	PASS
TC-SPC-IMR-014	LCL_I = 9.2124 ± 0.001 (imr_stable.csv)	LCL: 9.2124 ± 0.001	PASS
TC-SPC-IMR-015	UCL_MR = 1.0495 ± 0.001 (imr_stable.csv)	UCL_MR: 1.0495 ± 0.001	PASS
TC-SPC-IMR-016	--report → *_spc_imr_pv_report.docx or *_spc_imr_pv_report.html in ~/Downloads/	Exit 0; .docx or .html file found with recent mtime	PASS
TC-SPC-IMR-017	--report JSON sidecar *_spc_imr_pv_report_data.json (or	Exit 0; .docx or JSON found with recent mtime	PASS

	.docx accepted as evidence)		
TC-SPC-IMR-018	--report JSON: report_type == "pv", verdict_pass == True (IN CONTROL dataset)	Exit 0; if JSON present: report_type=="pv", verdict_pass is True	PASS

6.4 jrc_spc_xbar_r — 19/19 PASS

Script: jrc_spc_xbar_r.R — X-bar & R (X-bar and Range) chart. Test data: xbar_r_stable.csv (stable), xbar_r_ooc.csv (out-of-control).

TC ID	Description	Expected	Result
TC-SPC-XBR-001	Stable dataset — exit 0, X-bar and R sections and verdict present	Exit 0, X-bar, Range sections, verdict	PASS
TC-SPC-XBR-002	Stable data — IN CONTROL verdict	Exit 0, 'IN CONTROL' or 'STABLE'	PASS
TC-SPC-XBR-003	OOC dataset — exit 0, OUT OF CONTROL	Exit 0, 'OUT OF CONTROL' or 'SIGNALS'	PASS
TC-SPC-XBR-004	OOC dataset — subgroup sg09 flagged as violation	Exit 0, 'sg09' in output	PASS
TC-SPC-XBR-005	--ucl / --lcl flags accepted	Exit 0, user-specified UCL mentioned	PASS
TC-SPC-XBR-006	PNG written to ~/Downloads/	PNG present	PASS
TC-SPC-XBR-007	No arguments — non-zero exit, usage message	Exit ≠ 0, 'Usage'	PASS
TC-SPC-XBR-008	File not found — non-zero exit	Exit ≠ 0	PASS
TC-SPC-XBR-009	Missing column — non-zero exit	Exit ≠ 0	PASS
TC-SPC-XBR-010	Unbalanced subgroups — non-zero exit	Exit ≠ 0	PASS
TC-SPC-XBR-011	Subgroup size > 10 — non-zero exit, guidance message	Exit ≠ 0	PASS
TC-SPC-XBR-012	Bypass protection	Exit ≠ 0, 'RENV_PATHS_ROOT'	PASS
TC-SPC-XBR-013	Grand X-bar = 50.165 ± 0.001 (xbar_r_stable.csv)	Grand X-bar: 50.165 ± 0.001	PASS
TC-SPC-XBR-014	UCL_x = 50.9151 ± 0.001	UCL: 50.9151 ± 0.001	PASS
TC-SPC-XBR-015	LCL_x = 49.4149 ± 0.001	LCL: 49.4149 ± 0.001	PASS
TC-SPC-XBR-016	UCL_R = 2.7482 ± 0.001	UCL_R: 2.7482 ± 0.001	PASS
TC-SPC-XBR-017	--report → *_xbar_r_pv_report.docx or *_xbar_r_pv_report.html in ~/Downloads/	Exit 0; .docx or .html file found with recent mtime	PASS
TC-SPC-XBR-018	--report JSON sidecar *_xbar_r_pv_report_data.json (or .docx accepted as evidence)	Exit 0; .docx or JSON found with recent mtime	PASS
TC-SPC-XBR-019	--report JSON: report_type == "pv", verdict_pass == True (stable dataset)	Exit 0; if JSON present: report_type=="pv", verdict_pass is True	PASS

6.5 jrc_spc_xbar_s — 18/18 PASS

Script: jrc_spc_xbar_s.R — X-bar & S (X-bar and Standard Deviation) chart. Test data: xbar_s_stable.csv (stable), xbar_s_ooc.csv (out-of-control).

TC ID	Description	Expected	Result
TC-SPC-XBS-001	Stable dataset — exit 0, X-bar and S sections and verdict present	Exit 0, X-bar, Std Dev sections, verdict	PASS
TC-SPC-XBS-002	Stable data — IN CONTROL verdict	Exit 0, 'IN CONTROL' or 'STABLE'	PASS
TC-SPC-XBS-003	OOC dataset — exit 0, OUT OF	Exit 0, 'OUT OF CONTROL' or	PASS

	CONTROL	'SIGNALS'	
TC-SPC-XBS-004	OOc dataset — subgroup sg09 flagged as violation	Exit 0, 'sg09' in output	PASS
TC-SPC-XBS-005	--ucl / --lcl flags accepted	Exit 0, user-specified UCL mentioned	PASS
TC-SPC-XBS-006	PNG written to ~/Downloads/	PNG present	PASS
TC-SPC-XBS-007	No arguments — non-zero exit, usage message	Exit ≠ 0, 'Usage'	PASS
TC-SPC-XBS-008	File not found — non-zero exit	Exit ≠ 0	PASS
TC-SPC-XBS-009	Missing column — non-zero exit	Exit ≠ 0	PASS
TC-SPC-XBS-010	Unbalanced subgroups — non-zero exit	Exit ≠ 0	PASS
TC-SPC-XBS-011	Bypass protection	Exit ≠ 0, 'RENV_PATHS_ROOT'	PASS
TC-SPC-XBS-012	Grand X-bar = 100.060 ± 0.001 (xbar_s_stable.csv)	Grand X-bar: 100.060 ± 0.001	PASS
TC-SPC-XBS-013	UCL_x = 100.5807 ± 0.001	UCL: 100.5807 ± 0.001	PASS
TC-SPC-XBS-014	LCL_x = 99.5393 ± 0.001	LCL: 99.5393 ± 0.001	PASS
TC-SPC-XBS-015	UCL_s = 0.8598 ± 0.001	UCL_s: 0.8598 ± 0.001	PASS
TC-SPC-XBS-016	--report → *_xbar_s_pv_report.docx or *_xbar_s_pv_report.html in ~/Downloads/	Exit 0; .docx or .html file found with recent mtime	PASS
TC-SPC-XBS-017	--report JSON sidecar *_xbar_s_pv_report_data.json (or .docx accepted as evidence)	Exit 0; .docx or JSON found with recent mtime	PASS
TC-SPC-XBS-018	--report JSON: report_type == "pv", verdict_pass == True (stable dataset)	Exit 0; if JSON present: report_type=="pv", verdict_pass is True	PASS

6.6 jrc_spc_p — 16/16 PASS

Script: jrc_spc_p.R — P-chart (proportion nonconforming). Test data: p_stable.csv (stable), p_ooc.csv (out-of-control).

TC ID	Description	Expected	Result
TC-SPC-P-001	Stable dataset — exit 0, p-bar, control limits, verdict present	Exit 0, p-bar, UCL, verdict	PASS
TC-SPC-P-002	Stable data — IN CONTROL verdict	Exit 0, 'IN CONTROL' or 'STABLE'	PASS
TC-SPC-P-003	Known p_bar ≈ 0.041 in output	Exit 0, value near 0.041	PASS
TC-SPC-P-004	OOc dataset — exit 0, OUT OF CONTROL	Exit 0, 'OUT OF CONTROL' or 'SIGNALS'	PASS
TC-SPC-P-005	OOc dataset — subgroup 13 flagged as violation	Exit 0, '13' in output	PASS
TC-SPC-P-006	PNG written to ~/Downloads/	PNG present	PASS
TC-SPC-P-007	No arguments — non-zero exit, usage message	Exit ≠ 0, 'Usage'	PASS
TC-SPC-P-008	File not found — non-zero exit	Exit ≠ 0	PASS
TC-SPC-P-009	Missing column — non-zero exit	Exit ≠ 0	PASS
TC-SPC-P-010	defectives > n — non-zero exit	Exit ≠ 0	PASS
TC-SPC-P-011	Bypass protection	Exit ≠ 0, 'RENV_PATHS_ROOT'	PASS
TC-SPC-P-012	p-bar = 0.04160 ± 0.0001 (p_stable.csv)	p-bar: 0.04160 ± 0.0001	PASS
TC-SPC-P-013	UCL = 0.10150 ± 0.0010 (p_stable.csv)	UCL: 0.10150 ± 0.001	PASS
TC-SPC-P-014	--report → *_spc_p_pv_report.docx or *_spc_p_pv_report.html in	Exit 0; .docx or .html file found with recent mtime	PASS

	~/Downloads/		
TC-SPC-P-015	--report JSON sidcar *_spc_p_pv_report_data.json (or .docx accepted as evidence)	Exit 0; .docx or JSON found with recent mtime	PASS
TC-SPC-P-016	--report JSON: report_type == "pv", verdict_pass == True (stable dataset)	Exit 0; if JSON present: report_type=="pv", verdict_pass is True	PASS

6.7 jrc_spc_c — 12/12 PASS

Script: jrc_spc_c.R — C-chart (count of defects per unit). Test data: c_stable.csv (stable), c_ooc.csv (out-of-control).

TC ID	Description	Expected	Result
TC-SPC-C-001	Stable dataset — exit 0, c-bar, control limits, verdict present	Exit 0, c-bar, UCL, verdict	PASS
TC-SPC-C-002	Stable data — IN CONTROL verdict	Exit 0, 'IN CONTROL' or 'STABLE'	PASS
TC-SPC-C-003	Known c_bar $\approx$ 4.960 in output	Exit 0, value near 4.96	PASS
TC-SPC-C-004	OOC dataset — exit 0, OUT OF CONTROL	Exit 0, 'OUT OF CONTROL' or 'SIGNALS'	PASS
TC-SPC-C-005	OOC dataset — subgroup 12 flagged as violation	Exit 0, '12' in output	PASS
TC-SPC-C-006	PNG written to ~/Downloads/	PNG present	PASS
TC-SPC-C-007	No arguments — non-zero exit, usage message	Exit $\neq$ 0, 'Usage'	PASS
TC-SPC-C-008	File not found — non-zero exit	Exit $\neq$ 0	PASS
TC-SPC-C-009	Missing column — non-zero exit	Exit $\neq$ 0	PASS
TC-SPC-C-010	Bypass protection	Exit $\neq$ 0, 'RENV_PATHS_ROOT'	PASS
TC-SPC-C-011	c-bar = 4.960 ± 0.001 (c_stable.csv)	c-bar: 4.960 ± 0.001	PASS
TC-SPC-C-012	UCL = 11.641 ± 0.010 (c_stable.csv)	UCL: 11.641 ± 0.010	PASS

7. Deviations

No deviations were identified during OQ execution. All 83 test cases passed.

8. Requirements Traceability Matrix

The following table maps each User Requirement (UR) to the test cases that verify it. All 29 URs are fully covered by the 83 test cases executed in this OQ.

UR ID	Requirement Description	Verified By
UR-SPC-001	I-MR chart computes X-bar, MR-bar, UCL_I, LCL_I, UCL_MR	TC-SPC-IMR-001..004, 012..015
UR-SPC-002	I-MR user-specified limits accepted	TC-SPC-IMR-005
UR-SPC-003	I-MR chart PNG to ~/Downloads/	TC-SPC-IMR-006
UR-SPC-004	I-MR error handling (no args, file not found, missing col, too few obs)	TC-SPC-IMR-007..010
UR-SPC-005	I-MR bypass protection	TC-SPC-IMR-011
UR-SPC-006	X-bar & R computes grand X-bar, R-bar, UCL_x, LCL_x, UCL_R	TC-SPC-XBR-001..004, 013..016
UR-SPC-007	X-bar & R user-specified limits	TC-SPC-XBR-005

UR-SPC-008	X-bar & R PNG to ~/Downloads/	TC-SPC-XBR-006
UR-SPC-009	X-bar & R error handling (no args, file not found, missing col, unbalanced, size > 10)	TC-SPC-XBR-007..011
UR-SPC-010	X-bar & R bypass protection	TC-SPC-XBR-012
UR-SPC-011	X-bar & S computes grand X-bar, s-bar, UCL_x, LCL_x, UCL_s	TC-SPC-XBS-001..004, 012..015
UR-SPC-012	X-bar & S user-specified limits	TC-SPC-XBS-005
UR-SPC-013	X-bar & S PNG to ~/Downloads/	TC-SPC-XBS-006
UR-SPC-014	X-bar & S error handling (no args, file not found, missing col, unbalanced)	TC-SPC-XBS-007..010
UR-SPC-015	X-bar & S bypass protection	TC-SPC-XBS-011
UR-SPC-016	P-chart computes p-bar, UCL, LCL	TC-SPC-P-001..005, 012..013
UR-SPC-017	P-chart PNG to ~/Downloads/	TC-SPC-P-006
UR-SPC-018	P-chart error handling (no args, file not found, missing col, defectives > n)	TC-SPC-P-007..010
UR-SPC-019	P-chart bypass protection	TC-SPC-P-011
UR-SPC-020	C-chart computes c-bar, UCL, LCL	TC-SPC-C-001..005, 011..012
UR-SPC-021	C-chart PNG to ~/Downloads/	TC-SPC-C-006
UR-SPC-022	C-chart error handling (no args, file not found, missing col)	TC-SPC-C-007..009
UR-SPC-023	C-chart bypass protection	TC-SPC-C-010
UR-SPC-024	All charts — OOC observations flagged with rule number	TC-SPC-IMR-004, XBR-004, XBS-004, P-005, C-005
UR-SPC-025	All charts — stable data produces IN CONTROL verdict	TC-SPC-IMR-002, XBR-002, XBS-002, P-002, C-002
UR-SPC-026	All scripts — no-arguments usage message	TC-SPC-IMR-007, TC-SPC-XBR-007, TC-SPC-XBS-007, TC-SPC-P-007, TC-SPC-C-007
UR-SPC-027	All scripts — file not found error	TC-SPC-IMR-008, TC-SPC-XBR-008, TC-SPC-XBS-008, TC-SPC-P-008, TC-SPC-C-008
UR-SPC-028	All scripts — numeric correctness of control limits	TC-SPC-IMR-012..015, TC-SPC-XBR-013..016, TC-SPC-XBS-012..015, TC-SPC-P-012..013, TC-SPC-C-011..012
UR-SPC-029	jrc_spc_imr/xbar_r/xbar_s/p — --report PV report	TC-SPC-IMR-016..018, TC-SPC-XBR-017..019, TC-SPC-XBS-016..018, TC-SPC-P-014..016

9. OQ Conclusion

The Operational Qualification of the JR Validated Environment Statistical Process Control Module v2.0 is complete.

Result: PASS

All 83 automated test cases specified in JR-VP-SPC-001 v2.0 were executed on 2026-05-18 and 83 passed. Results are derived from the OQ evidence file listed below.

- jrc_spc_imr — I-MR (Individuals & Moving Range) Chart
- jrc_spc_xbar_r — X-bar & R (X-bar and Range) Chart
- jrc_spc_xbar_s — X-bar & S (X-bar and Standard Deviation) Chart
- jrc_spc_p — P-Chart (Proportion Nonconforming)
- jrc_spc_c — C-Chart (Count of Defects per Unit)

Evidence of test execution is archived at: spc_oq_execution_20260518T172901.txt

The Statistical Process Control module is released as version 2.0 of repos/spc/. Any subsequent change to the scripts, wrappers, or help files within repos/spc/ will require a change assessment and, if the change affects functionality covered by this OQ, re-execution of the affected test cases.